JAMA Ophthalmology | Original Investigation

Pickleball-Related Ocular Injuries Among Patients Presenting to Emergency Departments

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Supplemental content

IMPORTANCE Pickleball has experienced a surge in participation over the past decade, yet the epidemiology of pickleball-related ocular injuries remains unexplored.

OBJECTIVE To assess the incidence of and examine demographic factors associated with pickleball-related ocular injuries over the past 20 years.

DESIGN, SETTING, AND PARTICIPANTS This cross-sectional study used the National Electronic Injury Surveillance System (NEISS) database published by the US Consumer Product Safety Commission, which is based on a nationally representative probability sample of hospitals in the US and its territories. A population-based sample of cases was obtained by selecting patients who presented to NEISS-affiliated emergency departments with pickleball-related ocular injuries from 2005 to 2024. This resulted in a weighted estimate of ocular injuries nationally. Data were analyzed from October 6, 2024 to July 9, 2025.

EXPOSURE Participation in pickleball.

MAIN OUTCOMES AND MEASURES The main outcomes were the unweighted number and weighted number (percentage) of pickleball-related ocular injuries. Demographic information including age, sex, treatment year, and laterality of the injury were collected. The mechanisms of eye injuries and types of eye injuries were also assessed.

RESULTS Between 2005 and 2024, there were an estimated 3112 (unweighted n = 73) pickleball-related ocular injuries with an estimated 1262 (unweighted n = 33) ocular injuries occurring in 2024 alone. Based on a survey-weighted linear regression, the estimated incidence of pickleball-related eye injuries increased by 405 cases (95% CI, 288-522 cases; $P = .004$) annually from 2021 to 2024. There were more pickleball-related ocular injuries among patients aged 50 years or older than among patients younger than 50 years (difference, 39%; 95% CI, 6%-65%; $P = .03$). Severe eye injuries included retinal detachment, orbital fracture, globe trauma, and hyphema. Mechanisms of injury included direct hit by a pickleball, direct hit by a paddle, and falls.

CONCLUSIONS AND RELEVANCE This study found that pickleball-related eye injuries have increased at an alarming rate over the past 4 years as the sport continues to grow in popularity. Eye protection is not currently mandated for casual or professional play. Establishing standardized guidelines for eye protection is recommended to reduce the risk of ocular injuries among players.

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JAMA Ophthalmol. 2025;143(11):938-944. doi:10.1001/jamaophthalmol.2025.3577
Published online October 16, 2025.

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Invented in 1965 in Washington State, pickleball is a paddle sport that combines elements of tennis, badminton, and ping-pong and is played on a 6.0 × 13.2-m (20 × 44-ft) court with a perforated ball.[1] Pickleball has increased in popularity in the United States, with approximately 19.8 million players in the US as of 2024.[2-4] The sport's low-impact nature and straightforward rules contribute to its broad appeal and accessibility. As a result, the number of people engaged in the sport has increased 44% from 2023 to 2024 and an impressive 311% since 2020.[5]

Pickleball has also increased in player age diversity. Although often associated with older age groups, recent trends show a notable increase in younger player participation as well. In fact, the age range of 25 to 34 years has become the largest group, comprising 2.3 million participants.[6] The increasing popularity of pickleball across age groups highlights the growing need for public awareness of associated injuries. In 2023 alone, the estimated health care costs for all pickleball-related injuries, including eye injuries, ranged from $250 million to $500 million.[7] Many studies have explored injuries associated with racket sports, including tennis, badminton, and racquetball.[8-13] Although the literature on pickleball-related injuries has been expanding, to our knowledge, no epidemiologic studies have specifically focused on the incidence of ocular injuries.[14-19] This study analyzes pickleball-associated eye injuries among patients evaluated in emergency departments (EDs) using a national database in the United States between 2005 and 2024.

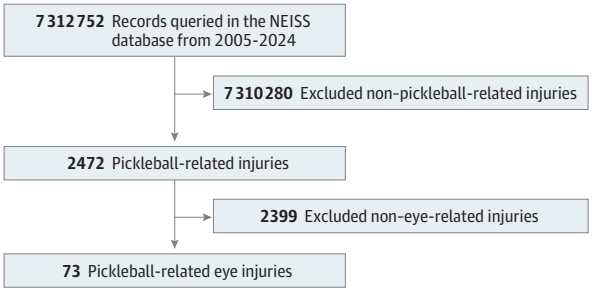
Key Points

Question What are the incidence of and demographic factors associated with pickleball-related ocular injuries over the past 20 years in the United States?

Findings In this cross-sectional study using the National Electronic Injury Surveillance System database between 2005 and 2024, the incidence of pickleball-related ocular injuries increased by an estimated 405 injuries per year from 2021 to 2024.

Meaning These findings suggest that pickleball-related ocular injuries have increased markedly over the past 4 years in conjunction with the sport's growing popularity, supporting the need to develop standardized guidelines for eye protection.

Figure 1. Pickleball-Related Eye Injury Case Selection Process From the National Electronic Injury Surveillance System (NEISS) Database



Methods

Database

This retrospective observational study used data obtained from the National Electronic Injury Surveillance System (NEISS), a government-funded database of deidentified patient information operated by the US Consumer Product Safety Commission. The Rutgers University Institutional Review Board determined the study to be nonhuman participants research and therefore exempt from review and informed consent. The NEISS collects data on consumer product-related injuries from a nationally representative probability sample of approximately 100 US EDs from more than 5000 hospitals nationwide. The sample is stratified by hospital size, geographic location, and age groups served. Collected data included patient demographic characteristics (age, sex, and race and ethnicity), primary diagnosis, injury location, and more. Each record also includes a narrative text field that provides descriptive details from ED documentation regarding the patient, history of present illness, and diagnosis. Additionally, each case is assigned a stratum, primary sampling unit, and survey weight to allow for calculation of a nationally representative estimate.[20] The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline was followed.[21]

Data

Data from January 1, 2005, to December 31, 2024, were queried, downloaded from the NEISS, and imported into Excel, version 16.92 (Microsoft) for review, filtering, and analysis. Pickleball-related eye injuries were identified with a computer-assisted search of the narrative text field for cases with the following: "pickleball" or "pickle ball" (n = 2443) and the common misspellings "pickelball" or "pickel ball" (n = 39) (**Figure 1**).[14] Body part codes 76 and 77 corresponding to "face" and "eye," respectively, were used for filtering. Nonocular injuries were excluded. Narratives of the remaining pickleball-related injuries were then reviewed to identify eye injuries not captured by body part codes. After appropriate case selection of pickleball-related eye injuries, the variables analyzed included treatment year, age, sex, mechanism of eye injury (fall related, hit by pickleball, hit by paddle, or not specified), type of eye injury, and the laterality of the injury (right or left). Severe eye injuries were defined by the authors as those that can result in permanent vision loss and/or require surgery. As per the US Consumer Product Safety Commission, national estimates were considered unstable and potentially unreliable when the estimate was less than 1200 cases, the number of cases was less than 20, or the coefficient of variation exceeded 33%.[22]

Statistical Analysis

All statistical analyses were performed using R, version 4.5.1 (R Project for Statistical Computing) with the survey package, which accounts for the complex sampling design of the NEISS database by incorporating case weights, stratification, and clustering to reduce sampling error and estimate

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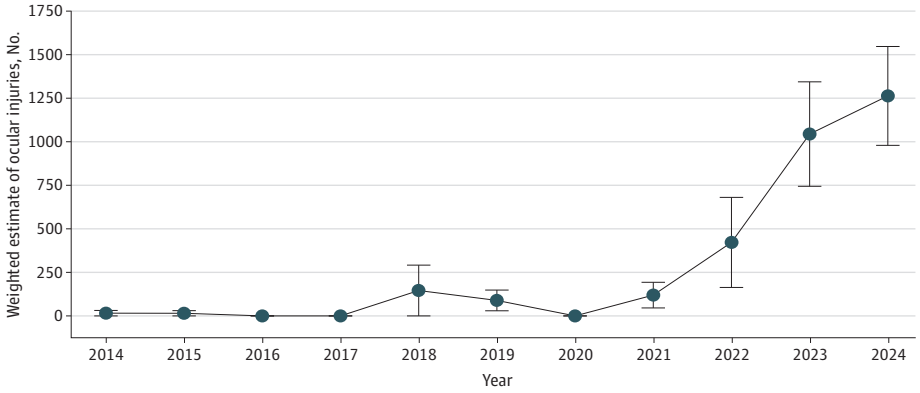
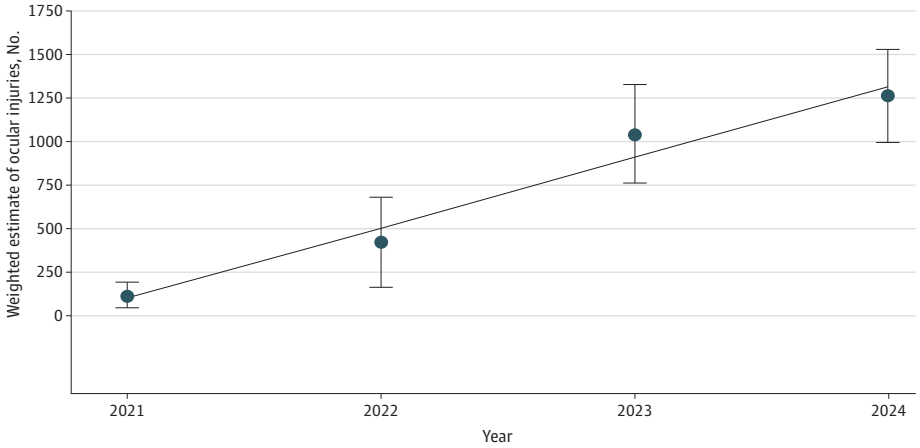
Figure 2. Weighted Estimates of Pickleball-Related Eye Injuries Presenting to Emergency Departments in the US Between 2014 and 2024



Error bars indicate 95% CIs.

Figure 3. Survey-Weighted Linear Regression of Pickleball-Related Ocular Injuries Between 2021 and 2024



Error bars indicate 95% CIs.

variance.[23] Weighted estimates were calculated using the sample weights assigned to each case by the NEISS.[20] A survey-weighted linear regression model was used to estimate the annual increase in the incidence of ocular injuries from 2021 to 2024. Survey-weighted binomial generalized linear models were used to compare weighted proportions between demographic subgroups. Comparisons were not performed when subgroup data did not meet reliability criteria outlined by the US Consumer Product Safety Commission. Wald tests were used to assess the statistical significance of regression coefficients, with SEs estimated by Taylor series linearization and degrees of freedom based on the survey design. $P < .05$ was considered statistically significant. All P values were 2-sided and were not adjusted for multiple analyses. Data were analyzed from October 6, 2024, to July 9, 2025.

Results

A total of 2472 records with pickleball-related injuries presenting to US EDs were identified between 2005 and 2024, resulting in a national estimate of 137 471 injuries. Of the 2472 records, 73 had documented eye injuries resulting in a weighted national estimate of 3112 eye injuries. Because no pickleball-related eye injuries were documented prior to 2014, those years were excluded from the analysis. Yearly estimates of pickleball-related eye injuries are depicted in **Figure 2**. In the most recent year (2024), there were an estimated 1262 (n = 33) eye injuries; 88% (95% CI, 78%-98%) of pickleball-related eye injuries reported occurred between 2022 and 2024. Based on a survey-weighted linear regression model, the estimated incidence of pickleball-related eye injuries increased by 405 cases (95% CI, 288-522 cases; $P = .004$) annually from 2021 to 2024 (**Figure 3**).

The demographic characteristics of the patients incurring pickleball-related eye injuries are presented in **Table 1**. The mean age was 53.9 years (95% CI, 46.8-61.1 years; median, 58 years [range, 6-80 years]). An estimated 1737 eye injuries (56%; unweighted n = 38) occurred among male participants, whereas an estimated 1375 injuries (44%; unweighted n = 35) occurred among female participants. There was no difference in the estimated number of eye injuries sustained by male and female participants (difference, 12%; 95% CI, −2% to 25%; $P = .11$). Among patients 60 years or older, there were an estimated 1619 pickleball-related eye injuries (52%; unweighted

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n = 32). An estimated 942 injuries (30%; unweighted n = 23) occurred among patients 30 to 59 years of age, and 551 (18%; unweighted n = 18) occurred among patients 0 to 29 years of age. There were more estimated pickleball-related eye injuries among patients aged 50 years or older (unweighted n = 43; weighted estimate [WE] = 2168 [70%]) compared with those younger than 50 years (unweighted n = 30; WE = 944 [30%]) (difference, 39%; 95% CI, 6%-65%; P = .03).

Raw counts and weighted estimates of ocular injury types are presented in **Table 2**. Documented ocular injuries included periocular lacerations (unweighted n = 25; WE = 1112 [35%]), corneal abrasions (unweighted n = 11; WE = 490 [16%]), iritis (unweighted n = 5; WE = 237 [8%]), periocular contusions (unweighted n = 4; WE = 149 [5%]), posterior vitreous detachment (unweighted n = 2; WE = 74 [2%]), subconjunctival hemorrhage (unweighted n = 3; WE = 162 [5%]), and traumatic mydriasis (unweighted n = 1; WE = 51 [2%]). The authors considered the following injuries to be severe ocular injuries: hyphema (unweighted n = 4; WE = 141 [5%]), globe trauma (unweighted n = 1; WE = 81 [3%]), retinal detachment (unweighted n = 2; WE = 79 [3%]), and orbital fracture (unweighted n = 3; WE = 49 [2%]). In 13 reported cases (WE = 516 [16%]), the type of eye injury was not specified.

The mechanisms of pickleball-related ocular injuries included direct hit by a ball (unweighted n = 34; WE = 1322 [43%]), falls (unweighted n = 18; WE = 884 [28%]), and direct hit by a paddle (unweighted n = 11; WE = 364 [12%]). There was no mechanism specified in 10 (WE = 542 [17%]) reported cases. The mean age of patients who sustained pickleball-related eye injuries was 68.2 years (95% CI, 65.2-71.2 years) for falls, 52.6 years (95% CI, 47.0-58.2 years) for ball-related injuries, and 30.5 years (95% CI, 16.6-44.4 years) for paddle-related injuries.

Table 1. Demographic Characteristics of Patients With Pickleball-Related Ocular Injuries

Characteristic	Unweighted No.	Weighted No. (%)
Age, y		
0-29	18	551 (18)
30-59	23	942 (30)
≥60	32	1619 (52)
Sex		
Male	38	1737 (56)
Female	35	1375 (44)
Laterality		
Right eye	28	1141 (37)
Left eye	31	1358 (44)
Not specified[a]	14	614 (20)

[a] Data not available.

Table 2. Diagnoses and Mechanisms of Injury Among Patients With Pickleball-Related Ocular Injuries

Variable	Unweighted No.	Weighted No. (%)
Diagnosis		
Periocular laceration	25	1112 (35)
Corneal abrasion	11	490 (16)
Iritis	5	237 (8)
Periocular contusion	4	149 (5)
Hyphema	4	141 (5)
SCH	3	162 (5)
Orbital fracture	3	49 (2)
Retinal detachment	2	79 (3)
PVD	2	74 (2)
Globe trauma	1	81 (3)
Traumatic mydriasis	1	51 (2)
Not specified[a]	13	516 (16)
Mechanism of injury		
Hit with ball	34	1322 (43)
Fall	18	884 (28)
Hit with paddle	11	364 (12)
Not specified[a]	10	542 (17)

Abbreviations: PVD, posterior vitreous detachment; SCH, subconjunctival hemorrhage.

[a] Data not available.

Discussion

This cross-sectional study revealed that the incidence of pickleball-related eye injuries presenting to US EDs increased annually from 2021 to 2024, paralleling the increase in pickleball participation. Recent surveys have demonstrated that this growth has been primarily driven by the influx of casual players, defined as those who play fewer than 8 times per year, rather than core players, who play 8 times per year or more.[5] Although the increase in ocular injuries may simply reflect the increasing number of players, it is also possible that casual players are more susceptible to injury due to limited experience, unfamiliarity with the game, or lower level of physical fitness. Further studies are needed to evaluate differences in the incidences and types of injuries between casual and core pickleball players across age groups.

Second, most pickleball-related eye injuries in our study occurred among older adults, with 70% of all eye injuries reported in players aged 50 years or older. The increased vulnerability of this population to sports-related injuries may be attributable to age-related decreases in muscle mass, bone density, and balance.[24] Thus, for players in this age group, potential strategies to prevent eye injuries include strength and balance training, stretching and warming up prior to play, and use of proper equipment, including court shoes. Eye protection should be emphasized, especially for players who use nonshatterproof spectacles, those with high myopia or a family history of retinal detachment, and players with cataracts who may experience blur or glare.[25] Although patients aged 50 years or older represented the majority of cases in our study, this proportion was lower than the 91% reported by Forrester[15] when analyzing all pickleball-related injuries between 2001 and 2017. This difference may reflect the increasing participation of younger individuals in the sport, with the average player age decreasing to 34.8 years as of 2024.[6]

The severity of ocular injuries varied widely among patients presenting to the ED. Less severe injuries included periocular lacerations, corneal abrasions, iritis, periocular

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contusions, posterior vitreous detachments, subconjunctival hemorrhage, and traumatic mydriasis. Although posterior vitreous detachments diagnosed following trauma may not be considered ocular injuries, they were included in this analysis given their association with retinal pathology.[26] Severe and vision-threatening ocular injuries included retinal detachments, globe trauma, orbital fractures, and hyphema. As the dataset provides no longitudinal or visual acuity data, it is possible that the true visual and financial burden of these injuries extends beyond what is reported due to subsequent costs of follow-up and/or intervention. For example, a study conducted at Helsinki University Eye Hospital between 2011 and 2012 reported that the 3-month follow-up cost associated with orbital fractures averaged €6050 (approximately $7100) per patient.[27] Further studies should evaluate the long-term visual outcomes of pickleball-associated eye injuries and the potential health system costs savings associated with injury prevention.

An estimated 43% of eye injuries in this study were attributable to direct pickleball impact, underscoring the role of ball speed and force in ocular injury risk. Biomechanical analyses have demonstrated that a pickleball traveling faster than 82 km/h (30 mph) can deform the cornea, retina, and vitreous and can exert zonular stress sufficient to dislocate the lens.[28] This velocity threshold has become increasingly relevant due to recent advancements in paddle technology. Modern paddles, incorporating materials such as ethylene-vinyl acetate foam, can now generate exit velocities approaching 96 km/h (60 mph).[29] Given that players are often positioned 4.2 m (14 ft) apart at the "no-volley" or "kitchen" line, these high speeds leave players with limited time to react and avoid impact.[7] Understanding how paddle characteristics affect ocular injury risk may help inform future safety guidelines and recommendations.

This study adds to the growing body of documented pickleball-related ocular injuries leading to potential visual compromise.[30-32] Atkinson et al[30] reported 2 patients who presented with retinal tears after direct hit to the eye with a pickleball, 1 of whom had a localized retinal detachment and vitreous hemorrhage. Huang and Greven[31] reported 2 patients who sustained crystalline lens subluxation after being struck by a pickleball and subsequently required scleral-fixated intraocular lens implantation. Boopathiraj et al[32] described a case of a patient hit with a pickleball who presented with a hyphema and intraocular pressure of 62 mm Hg. None of the reported patients were wearing eye protection at the time of injury.

Despite the risks of eye injury, eye protection is not mandated for casual or professional pickleball players. Most recently in March 2024, USA Pickleball, the governing body for pickleball in the United States, disapproved of a rule change that would mandate that players wear eye protection in USA Pickleball tournaments, citing challenges with enforcement. Although recommendations were made for USA Pickleball to develop a comprehensive policy on eye protection, such guidance has yet to be issued.[33] Furthermore, many pickleball clubs and public recreation areas do not require that players wear eye protection.[34]

In the absence of formal guidelines for eye protection in pickleball, the American Academy of Ophthalmology published 2 recommendations in 2024. They suggest that eyewear should meet the American Society for Testing and Materials F3164 guidelines, which is currently the standard for most racket sports.[34,35] These guidelines outline criteria for safe designs, optics, and materials for eye protection, including for players who wear corrective spectacles.[36] However, because these standards are neither mandated nor widely promoted by mainstream pickleball organizations, it is difficult to monitor protective eyewear compliance. Furthermore, many retailers may not disclose the materials used in their frames and lenses, and whether their products meet recommended standards. Adopting and publicizing a universal standard would simplify the purchasing process and ensure players have access to the most effective eye protection.

To contextualize our findings, we compared the annual number of pickleball-related ocular injuries with ocular injury rates reported in other racket and nonracket sports. In contrast to the increasing annual incidence of pickleball-related ocular injuries, tennis-related ocular injuries decreased from 2000 to 2019, averaging approximately 800 per year and decreasing to an estimated 450 eye injuries in 2019.[37] Baseball-related ocular injuries followed a similar pattern, with the estimated incidence decreasing annually from 2011 to 2020 to an estimated 200 ocular injuries in 2020.[38] Although a recent study on basketball-related eye injuries reported that an estimated 3000 injuries occurred in 2021, the annual incidence decreased by an estimated 333 cases per year between 2012 and 2021.[39] Thus, compared with tennis, baseball, and basketball, pickleball-related eye injuries may represent an increasing burden of sports-related ocular trauma. Efforts to curtail this increasing trend by increasing awareness and developing standardized guidelines for eye protection should be considered.

Limitations

This study has several limitations. First, the sample size was small, with only 73 reported cases identified over the 20-year study period. Because the NEISS data are derived from a sample of approximately 100 hospitals using probability sampling, the findings may not be truly representative of the more than 5000 hospitals in the United States with EDs.[40] Furthermore, pickleball-related injuries may not have been recognized in earlier years of the database given the sport's lower popularity during that time period and the absence of a specific NEISS product code. The authors attempted to capture all pickleball-related injuries by searching among narratives and removing product codes from the database query. However, because the narrative text field in the NEISS was expanded from 142 to 400 characters in 2019, it is possible that cases occurring before 2019 were not adequately documented.[22] The analysis was also limited to patients who presented to EDs, excluding those who sought care in ophthalmology offices or other outpatient settings. Therefore, the true number of pickleball-related ocular injuries is likely underestimated by this study, with potential bias toward more severe cases. Moreover, more subtle diagnoses, such as traumatic macular hole, may have been missed given the limited ophthalmic resources in most EDs. Due to the limited detail available within the database, it is unclear

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whether diagnoses were made by ophthalmologists or by other medical professionals.

Conclusions

This cross-sectional study found that the annual incidence of pickleball-related ocular injuries increased substantially from 2021 to 2024, with an estimated 1262 cases occurring in 2024 alone. Most affected individuals were 50 years of age or older. Severe ocular injuries included retinal detachment, globe trauma, orbital fracture, and hyphema. Currently, no official guidelines exist regarding eye protection in pickleball. Increasing awareness for age-specific risk factors and establishing standardized recommendations for eye protection may help reduce injury rates and prevent vision loss.

ARTICLE INFORMATION

Accepted for Publication: July 13, 2025.

Published Online: October 16, 2025. doi:10.1001/jamaophthalmol.2025.3577

Author Contributions: Drs Lacher and Tsui had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis.
Concept and design: Lacher, Tsui.
Acquisition, analysis, or interpretation of data: All authors.
Drafting of the manuscript: All authors.
Critical review of the manuscript for important intellectual content: All authors.
Statistical analysis: Lacher, Koc.
Administrative, technical, or material support: Tsui.
Supervision: Tsui.

Conflict of Interest Disclosures: Dr Tsui reported receiving consulting fees from AbbVie Inc and EyePoint and grants and unrelated research funding from Johnson & Johnson Surgical Vision outside the submitted work. No other disclosures were reported.

Meeting Presentation: This paper was presented at the Summer Symposium Meeting of the Women in Ophthalmology; August 9, 2025; Amelia Island, Florida.

Data Sharing Statement: See the Supplement.

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Invited Commentary

Benefits and Limitations of Using Population Databases for Eye Research—More Than Meets the Eye

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Compared with single or multicenter studies, health-related data repositories increase the scale, scope, and power of research aimed at identifying trends in ocular conditions across demographic groups, geographic regions, and time periods. Research using such databases has numerous advantages; however, it is important to also understand its limitations.

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As a case example, we highlight work in this issue of *JAMA Ophthalmology* from Lacher et al,[1] who explored trends in pickleball-related eye injuries using 2005 to 2024 data from the Consumer Product Safety Commission's National Electronic Injury Surveillance System (NEISS).[2] The NEISS contains coded entries of product-related consumer injuries derived from a sample of 100 hospital-based emergency departments in the US. Data from this probability sample, which represents approximately 2% of more than 5000 US emergency departments, can be used to generate weighted estimates of national incidence rates using a variety of statistical methods.

In addition to the NEISS, several other patient-related databases represent important resources to explore epidemiologic patterns, associations, and disparities in eye health. Some databases, such as the NEISS and the Nationwide Emergency Department Sample,[3] are based on emergency department data, whereas others include data from outpatient practices (including community-based or private practices, as well as academic or university-based clinics) or from broader surveillance systems, surveys, or administrative data. Examples include the American Academy of Ophthalmology's Intelligent Research in Sight (IRIS)[4] Registry, the University of Michigan's Sight Outcomes Research Collaborative (SOURCE),[5] the Center for Disease Control and Prevention's Vision Health Surveillance System,[6] the National Ambulatory Medical Care Survey,[7] and the Centers for Medicare & Medicaid Services Transformed Medicaid Statistical Information System[8] Analytic Files.

By including diverse and representative populations, research using these databases may produce results that are more generalizable than single-center studies. These databases also facilitate large-scale observational studies that could otherwise be logistically and financially challenging. The results of studies using them might support the generation of new hypotheses for prospective studies, guide comparative effectiveness research, or inform health policy measures and public safety regulation. The latter is the case of the study by Lacher et al,[1] which aims to raise awareness of pickleball-related eye injuries to promote the use of protective equipment to safeguard sight.

Selecting the most appropriate dataset for each study's objectives is critical. As mentioned, the NEISS, for example, is focused on consumer product safety. However, the NEISS is limited by the number of hospitals sampled, the lack of follow-up data for each patient, and the restriction to emergency department visits. Other databases, however, may include data on visit types, diagnoses, procedures, prescriptions, or billing summaries. They also may include cases that were treated outside of an emergency department setting. Others also may include information pertaining to nonocular comorbid conditions, laboratory values, or ocular or other radiologic images, potentially facilitating the examination of associations between ocular conditions and broader clinical phenotypes. Finally, some include longitudinal data that may allow for the assessment of practice patterns, disease progression, or responses to therapies.

As with all sources of data, registries have inherent limitations, including potential challenges with insufficient, inaccurate, incomplete, and missing data. Human error and system-level gaps stemming from high demands on limited resources also can affect data quality. For example, because diagnostic and billing codes are not specifically intended for research, coding practices can lead to misclassification bias in data sources based on billing codes. Manual entry by coders

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